Guidelines for Mentees:

- **This program is a collaborative self-learning journey aimed at enhancing your logical thinking, understanding of data structures, basic programming, and problem-solving skills.**

- Each mentee will be assigned mentors in groups of 5. Mentors will create WhatsApp groups to stay connected and provide guidance.

- Weekly sync-ups will be conducted by Mentors to review your progress, assist with assignment completion, and address any doubts or challenges you may face.

- **Mentors are here to guide and support you throughout the program. Feel free to reach out whenever you need help or clarification.**

- Avoid using AI tools for completing assignments. The focus is on building your skills and confidence through your own efforts.

- **Mentees must adhere to strict deadlines.**

- Code of Conduct: Follow the guidelines of the program by adhering to deadlines, maintaining discipline, and actively participating to make the most of this learning program.

- Mentors will provide weekly constructive feedback and ratings to help you improve. Use feedback positively to enhance your skills.

- For any issues related to assignment submission or other concerns, kindly inform your mentor at least two days in advance.

Session Guidelines:

- **It is mandatory to attend each session**. In case of an emergency or prior commitment, please inform both your Mentor and HR in advance.
- **Join the meeting 10 minutes early with a stable internet connection, and ensure your video remains on at all times.**
- Each session will focus on a pre-defined topic. You are required to:
  - ❖ Learn the topic beforehand.
  - ❖ Implement that Data Structure in your Repository.

- ❖ Solve the Given Hacker Rank assignments.

Presentation Guidelines:

- Each group of mentors will be assigned a topic. PPTs will be prepared in groups of 5 and reviewed by mentors. Ensure your presentations are:
  - ❖ Interactive.
  - ❖ Include real-life examples.
  - ❖ Highlight the need, usage, and importance of the data structure (**without explaining the basics**).
  - ❖ Incorporate MCQ-based questions for interactive sessions.
  - ❖ Open to innovative ideas.

Assessments:

- There will be Three assessments during this program. It is mandatory to attend these assessments at the office campus (Jaipur or Bangalore). To Qualify from this program, you must:
  - ❖ Clear at least 2 out of 3 assessments.
  - ❖ Score a minimum of 3 out of 5 in weekly progress ratings.

Set-up for Hacker Rank:

- ❖ Sign-In to this contest: https://www.hackerrank.com/intimetec-kp-2025-batch with same HackerRankId/Credentials shared in google form before.
- ❖ Solve the First Challenge: "Solve Me First".

Pre-Kalpavriksha Assignment Review:

- Your mentors will review your assignments and provide comments on your pull requests (PRs). Please Ensure to address the comments and refactor your code before the next session.

Course Curriculum (Self-Learning):

Week 1: **Getting Started with C Language**

- Why C?
- The `main ()` function, Compilation, and Running the Program
- Variables, Data Types, Constants
- Input/Output in Detail: `printf()`, `scanf()`, etc.
- Arithmetic Operators
- Control Flow (if-else, switch-case)
- Simple Pattern Programs
- Loops (`for`, `while`, `do-while`) & Jump Statements (`break`, `continue`, `goto`)
- Functions: Arguments, Return Types, Recursion
- Structures
- Scope Rules

Week 2: **Functions, Pointers, Arrays and Strings**

- Why Pointers? Pointers & Types of Pointers
- Call by Value vs. Call by address
- Why Arrays? Arrays and Their Operations
- Character Arrays
- Input/Output with Arrays, Character Arrays, and Pointers
- Pointer Arithmetic
- Arrays with Addresses and Pointers
- Character Arrays as Pointers

Week 3: **Memory Management & Multidimensional Array**

- Memory Management: Stack vs. Heap
- Why DMA? Dynamic Memory Allocation (`malloc`, `calloc`, `free`)
- How Variables Are Stored and Accessed
- When to Use Stack vs Heap

- Leak, segmentation fault
- Multidimensional Arrays (2D and 3D)
- String Manipulation & Operations on Strings

Data Structures

Week 4: **Starting with Data Structures (Arrays, Linked Lists)**

- Why Linked Lists over Array?
- Implement Linked Lists from Scratch in C
- Perform CRUD Operations on the Linked List
- Practical Applications of Different Types of Linked Lists

Week 5: **Searching, Sorting and Recursion**

- Why Searching, Sorting and Recursion?
- Types of Searching and Sorting algorithms
- Implement Searching and Sorting on Arrays and Linked Lists

Week 6: **Stack, Queue and HashMap**

- Why Stack, Queue and HashMap?
- Real-life Applications of stack and Queue
- Collision Handling in HashMap / Hash Function
- Implement Stack, Queue, and HashMap Using Linked Lists
- Perform CRUD Operations on the Implemented Data Structures

Operating Systems

Week 8: **Starting with Operating System**

- Functions of an Operating System: Process Management, Memory Management, I/O Handling
- Thread vs Process: Difference and Use cases
- Fork () and exec ()
- Critical section, race condition, thread synchronization (semaphores and mutex)

Week 9: **Operating System**

- Multithreading
- Interrupts vs signals
- Thread and Process synchronization

Sessions will resume from 23rd December 2024, every Monday at 3 p.m.